Relationships between variables

1 State whether there is likely to be a relationship between the following pairs of variables:

a Study time and test score.

b Eye color and IQ.

c Typing speed and dancing ability.

d Driving time and gas used.

2 For each pair of variables, state whether one variable has a direct effect on the other. If yes, state which variable affects the other.

a Arm length (cm), Weight (kg)

b Temperature ($^\circ\mathrm{C}$), Distance from the equator (km)

c Weekly income, Number of friends

d Time of travel (minutes), Distance covered (m)

e Age (years), Time spent watching television

f Time spent working, Wages earned

g Time spent to finish a novel, Number of pages

h Marital status, Film preference

i Height (cm), Number of children

j Hair Length, Gender

k Number of pets, Time spent caring for pets

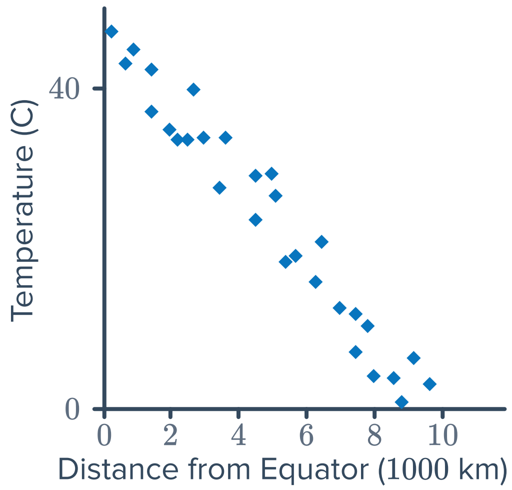
l Time spent training, Athletic performance

3 Scientists conducted a study where each person was asked to read a paragraph and then recount as much information as they can remember. They found that the longer the paragraph, the less information each person could retain.
If the length of the paragraph were plotted (on the horizontal axis) against the amount of information retained (on the vertical axis), would the correlation be negative or positive?

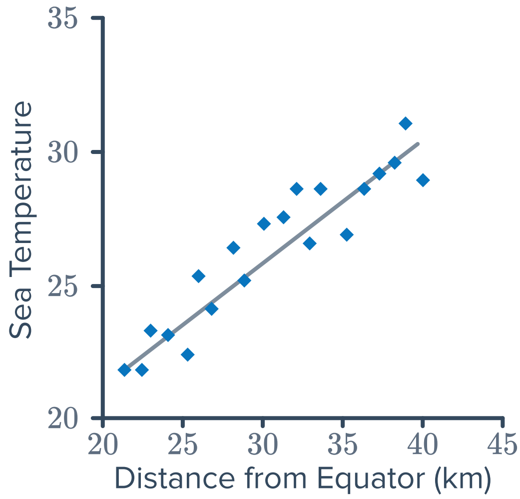
Scatter plots

4 Determine whether the following graphs show a negative relationship between the two variables:

a



b



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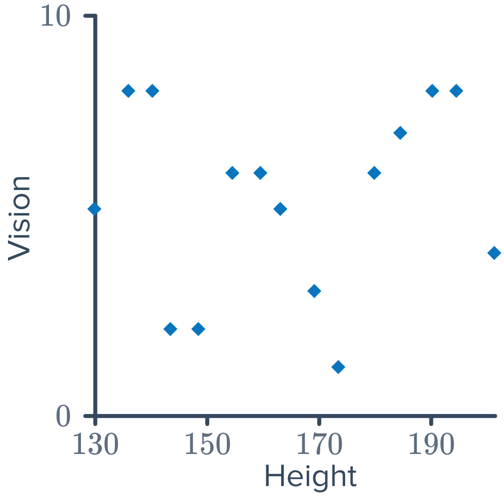
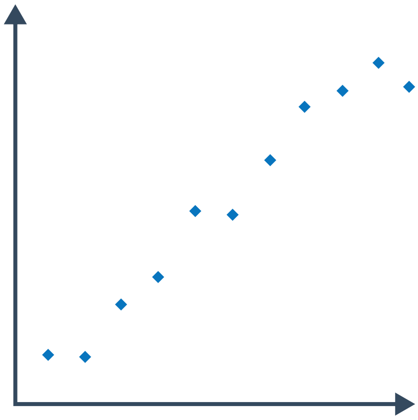


5 Construct a scatter plot for the set of data in the table.

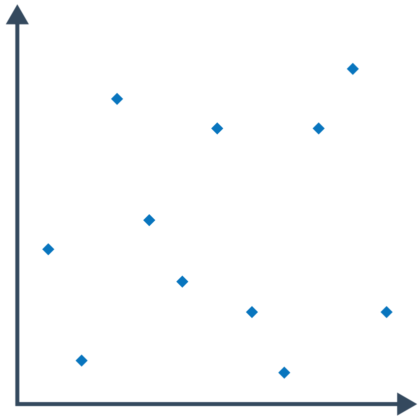
x	1	3	5	7	9
y	3	7	11	15	19

6 For each of the following scatter plots, de[redacted] the correlation in terms of strength and direction:

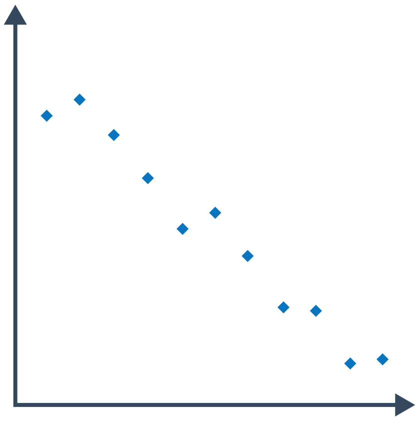
a



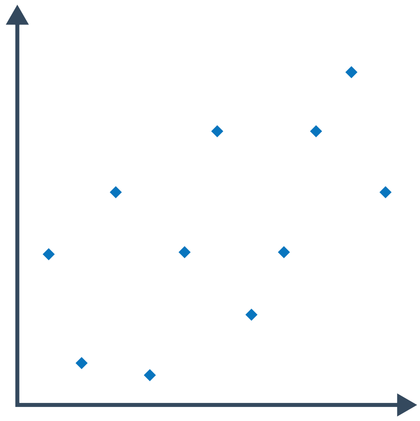
b



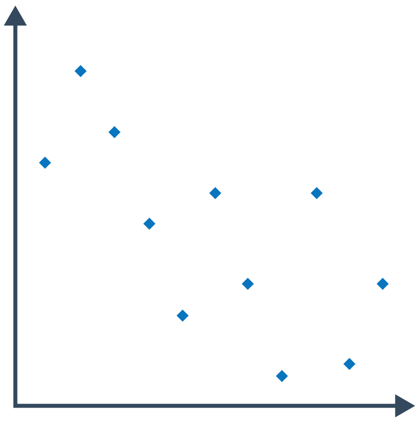
c



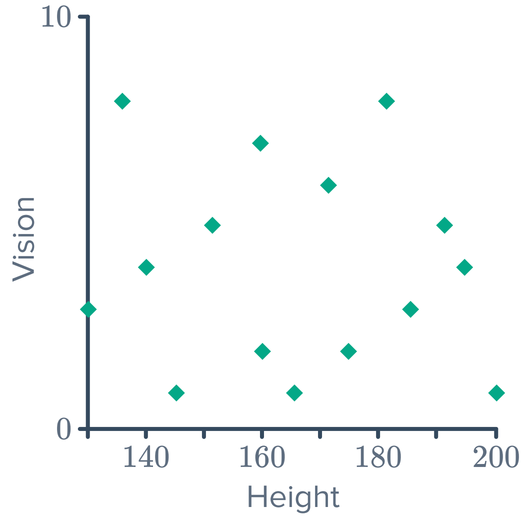
d



e



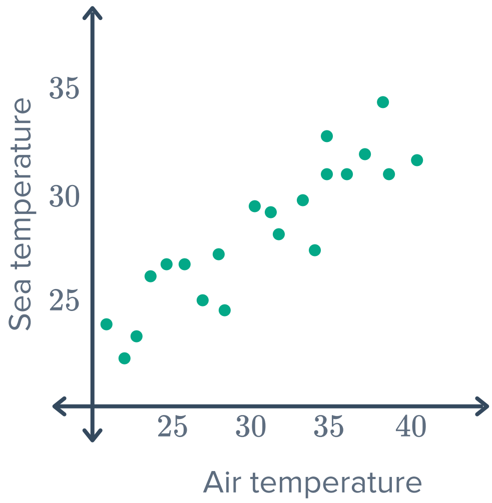
7 According the scatter plot shown, describe the relationship between someone's vision and their height.



8 The scatter plot shows the relationship between air and sea temperature:

a Describe the relationship between air and sea temperature.

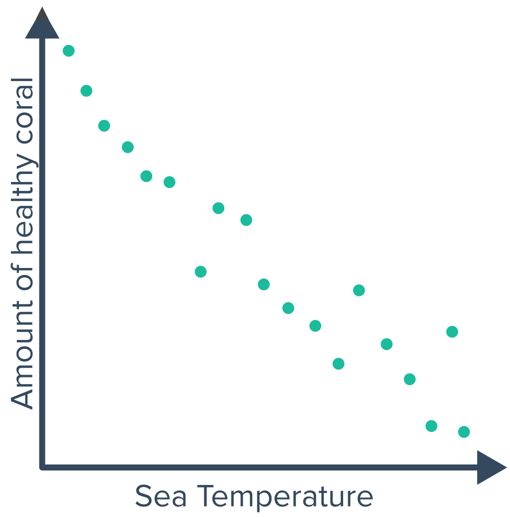
b Describe the correlation between air and sea temperature in terms of strength, direction and form.

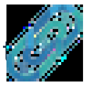


9 The scatter plot shows the relationship between sea temperature and the amount of healthy coral:

a Which variable is the dependent variable?

b Which variable is the independent variable?



10 The following table shows the number of  accidents associated with a sample of drivers of different age groups.

Age	20	25	30	35	40	45	50	55	60	65
Accidents	41	44	39	34	30	25	22	18	19	17

a Construct a scatter plot for this data.

b Describe the correlation between a person's age and the number of accidents.

c Does this data set contain any outliers?

11 The scores of 12 students in Math and Sport were recorded in the following table:

a Construct a scatter plot for the students' scores in math vs their scores in sports.

b Describe the correlation between students' scores in math and scores in sport.

c Which student's scores appear to represent an outlier?

Student	Score in Math	Score in Sports
1	63	44
2	92	74
3	60	52
4	79	70
5	88	67
6	81	60
7	61	73
8	91	86
9	72	84
10	42	93
11	66	57
12	92	92

12 The following table shows the average IQ of a random group of people against their height:

Height (cm)	140	145	150	155	160	165	170	175	180	185
IQ	103	95	98	111	85	89	108	145	110	93

a Construct a scatter plot to represent this data:

b Describe the relationship between IQ and height.

c How tall is the person who appears to be an outlier?

13 A researcher is studying the relationship between the number of passers-by present in a situation and the time taken, in seconds, until a stranger in an emergency receives help from a passer-by. The data is recorded in the table below:

a Construct a scatter plot to represent the data in the table.

b Comment on the relationship between the number of passers-by and the time until assistance is offered by a passer-by to a person in an emergency.

c Describe the correlation between the number of passers-by and the time until assistance is offered by a passer-by to a person in an emergency.

Passers-by	Time until offered help
1	8
2	19
3	26
4	37
5	51
6	65

14 The following table shows the scores of 12 students in English and French:

a Construct a scatter plot for English score vs French score.

b Describe the correlation between students' English and French scores.

c Which student's scores appear to represent an outlier?

Student	English	French
1	85	89
2	71	71
3	57	56
4	60	62
5	79	86
6	76	75
7	71	77
8	91	86
9	50	90
10	49	47
11	66	67
12	92	92

15 The table lists the time taken to sprint 400 meters by runners who all ran in different temperatures as part of a study:

a Construct a scatter plot to represent the data in the table.

b How many runners were tested in the study?

c Describe the correlation between temperature and sprint time for the data.

d Which data point represents an outlier?

Temperature ($^{\circ}$C)	Time (sec)
5	60
2	67
10	48
8	69
1	65
7	49
6	57
4	53
3	59
9	52

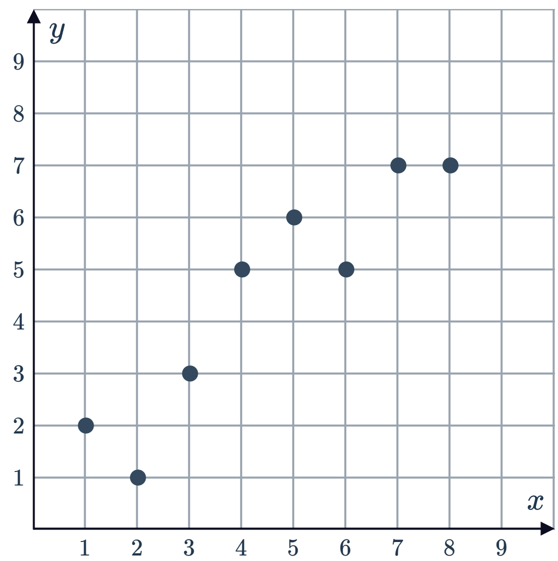
Lines of best fit

16 The following scatter plot shows the data for two variables, x and y:

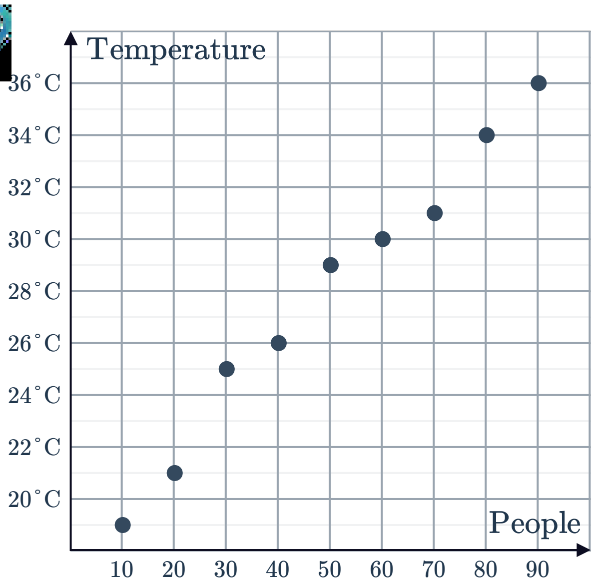
a Sketch the line of best fit for this data.

b Use your line of best fit to estimate the value of y when:

i $x = 4.5$ ii $x = 9$



17 The following scatter plot graphs data for the number of people in a room and the room temperature collected by a researcher.



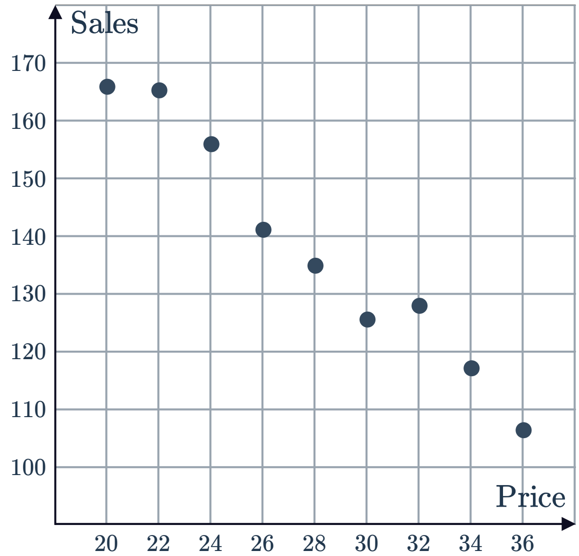
a Sketch the line of best fit for this data.

b Use your line of best fit to estimate the room temperature when there are:

i 35 people in the room

ii 100 people in the room

18 The following scatter plot graphs data for the number of copies of a particular book sold at various prices:

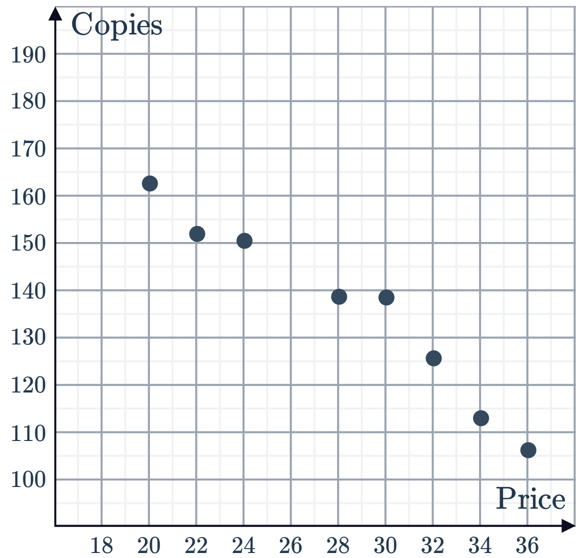


a Sketch the line of best fit for this data.

b Use your line of best fit to find the number of books that will be sold when the price is:

i $\$33$ ii $\$18$

19 The following scatter plot graphs data for the number of copies of a particular book sold at various prices.



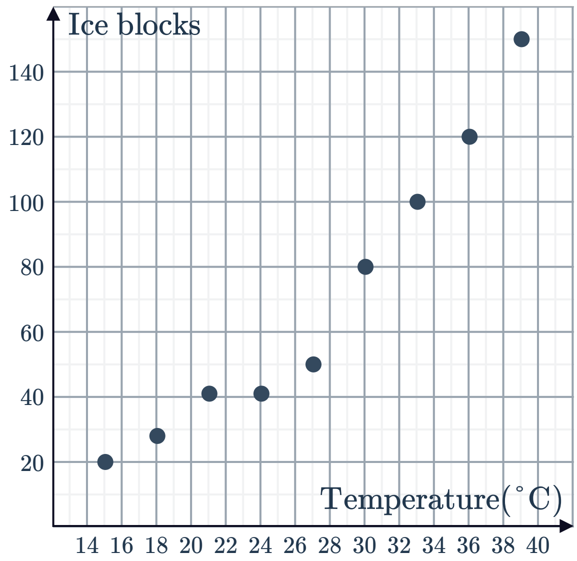
a Sketch the line of best fit for this data.

b Use the line of best fit to find the number of books that will be sold when the price is:

i $\$33$ ii $\$18$

c Is the relationship between the price of the book and the number of copies sold positive or negative?

20 The following scatter plot graphs data for the number of ice blocks sold at a shop on days with different temperatures.



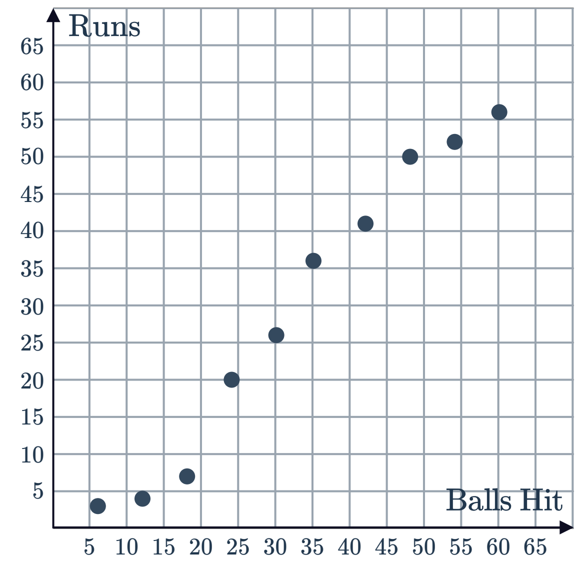
a Sketch the line of best fit for this data.

b Use your answer line of best fit to estimate the number of ice blocks that will be sold on a:

i $31^\circ\mathrm{C}$ day   **ii** $42^\circ\mathrm{C}$ day

c Does the number of ice blocks sold increase or decrease as the temperature increases?

21 The following scatter plot graphs data for the number of balls hit and the number of runs scored by a batsman:



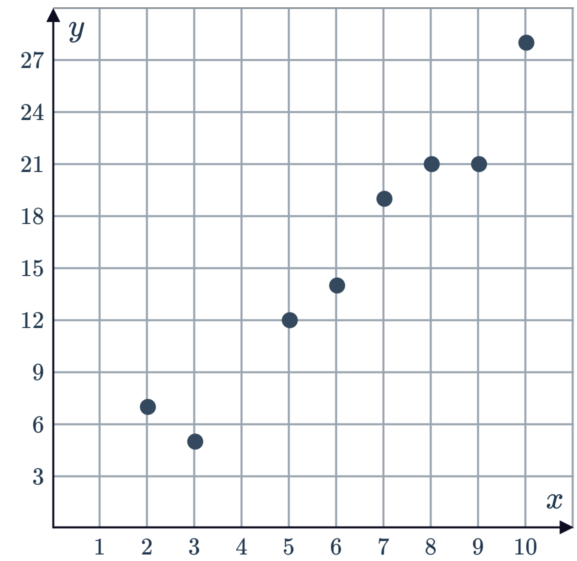
a Sketch the line of best fit for this data.

b Use the line of best fit to estimate the number of runs scored by the batsman after hitting:

i 27 balls   **ii** 66 balls

c Is the relationship between the two variables positive or negative?

22 Consider the following scatter plot:



a Is the relationship between the x and y variables positive or negative?

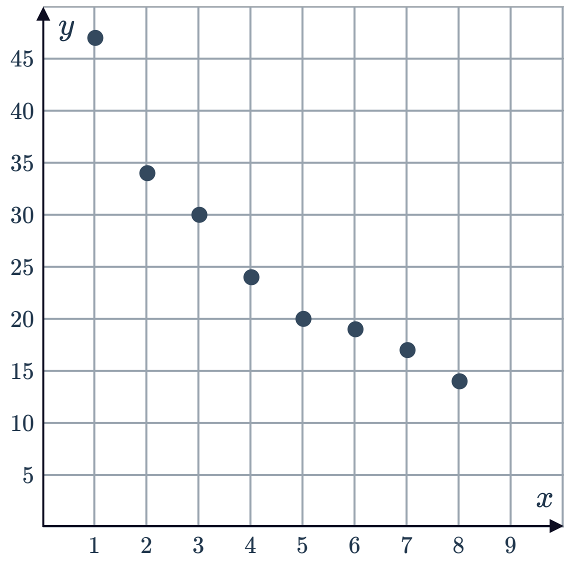
b Sketch the line of best fit for this data.

c Which of the following could be the equation for the line of best fit:

A $y = 2 - 3x$ **B** $y = 3x + 2$

C $y = -3x - 2$ **D** $y = 3x - 2$

23 Consider the following scatter plot:



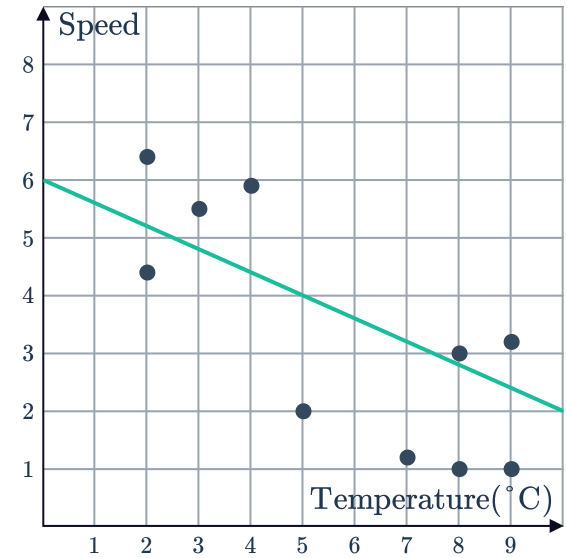
a Is the relationship between the x and y variables positive or negative?

b Which of the following could be the equation for the line of best fit:

A $y = -4x - 4$ **B** $y = 44 + 4x$

C $y = -4x + 44$ **D** $y = 4x - 4$

24 The average monthly temperature and the average wind speed, in knots, in a particular location was plotted over several months. The graph shows the points for each month's data and their line of best fit:
Use the line of best fit to approximate the wind speed on a day when the temperature is $5^\circ\mathrm{C}$.



25 A dam used to supply water to the neighboring town had the following data recorded for its volume over a number of months.

Month (x)	1	2	3	4
Volume, in billion of liters (y)	116	106	104	92

a Plot the points on a coordinate plane.

b Sketch the graph of the line of best fit.

26 Jordano measures his heart rate at various [redacted]s while running. The data is shown below in the table:

Time (minutes)	0	2	4	6	8	10
Heart Rate (BPM)	58	54	58	67	67	74

a Plot the data and sketch the line of best fit.

b Which data point is closest to the line of best fit?

c Which data point is furthest from the line of best fit?

27 The number of fish in a river is measured over a five year period:

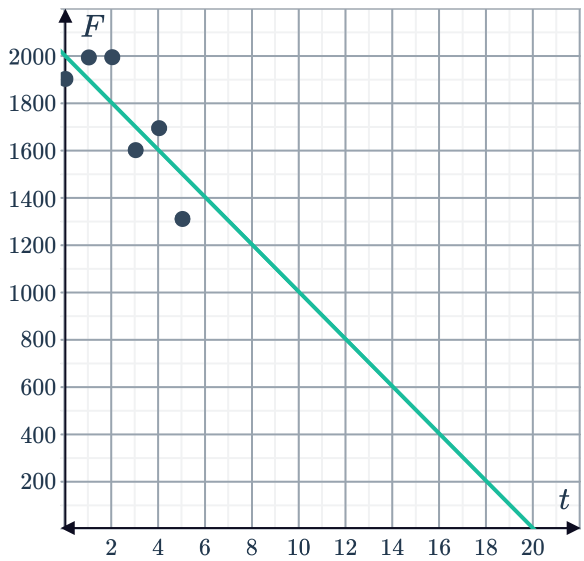
Time in years (t)	0	1	2	3	4	5
Number of fish (F)	1903	1994	1995	1602	1695	1311

The data has been graphed along with a line of best fit:

a Predict the number of years until there are no fish left in the river.

b Predict the number of fish remaining in the river after 7 years.

c According to the line of best fit, how many years are there until there are 900 fish left in the river?



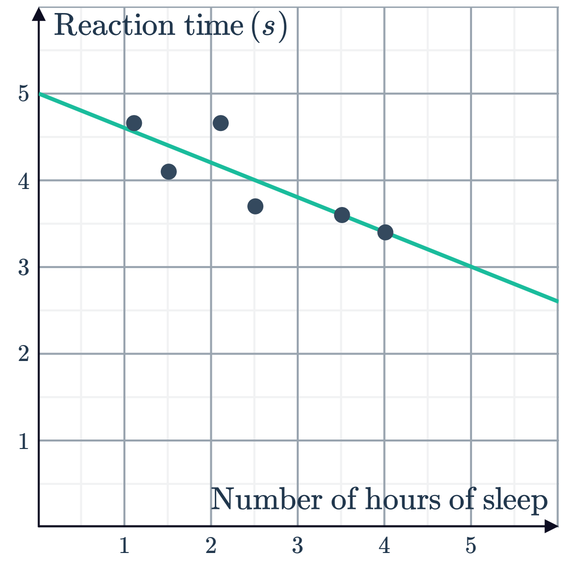
28 Scientists conducted a study to see people's reaction times after they've had different amounts of sleep. The results are recorded in the table:

Number of hours of sleep (x)	1.1	1.5	2.1	2.5	3.5	4
Reaction time in seconds (y)	4.66	4.1	4.66	3.7	3.6	3.4

The data has been graphed along with a line of best fit:

a Predict the reaction time for someone who has slept for 5 hours.

b Predict the number of hours someone sleeps if they have a reaction time of 4 seconds.



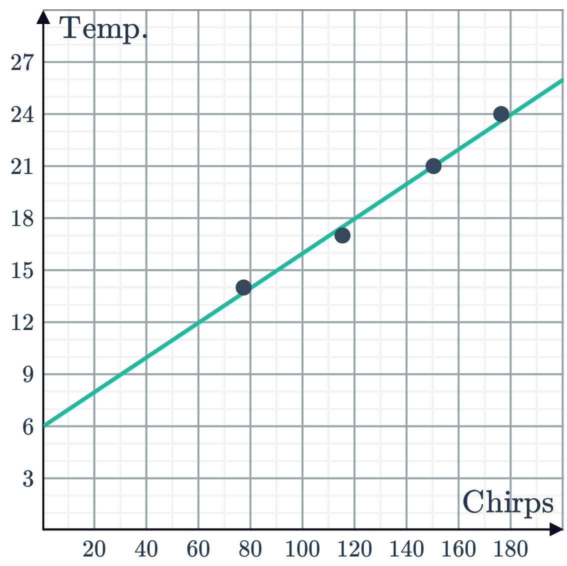
29 Chirping crickets can be an excellent indication on how hot or cool it is outside. Different species of crickets have different chirping rates but for a particular species the following data was recorded:

Number of chirps per minute	77	115	150	176
Temperature ($^\circ$C)	14	17	21	24

a What is the temperature when the crickets make 140 chirps each minute?

b How many chirps per minute will the crickets make if the temperature is 27?

c How many chirps are the crickets making each minute if the temperature is 19°C?



30 A plane's altitude (A) is measured at sever[redacted]es (t) during its descent:

Time (t seconds)	0	200	400	1700
Altitude (A meters)	9000	7815	7092	593

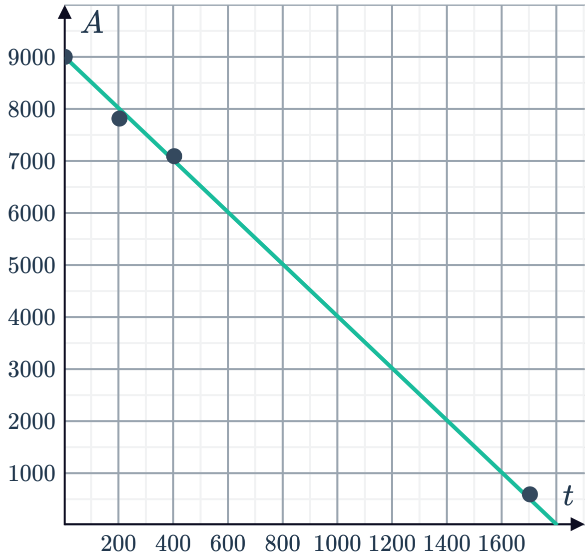
The data has been graphed along with a line of best fit:

a Predict the altitude of the plane 100 seconds into the descent.

b Predict the altitude of the plane 500 seconds into the descent.

c For how many seconds has the plane been descending when it is at an altitude of 7500 meters?

d How many seconds did the plane take to descend to the ground?

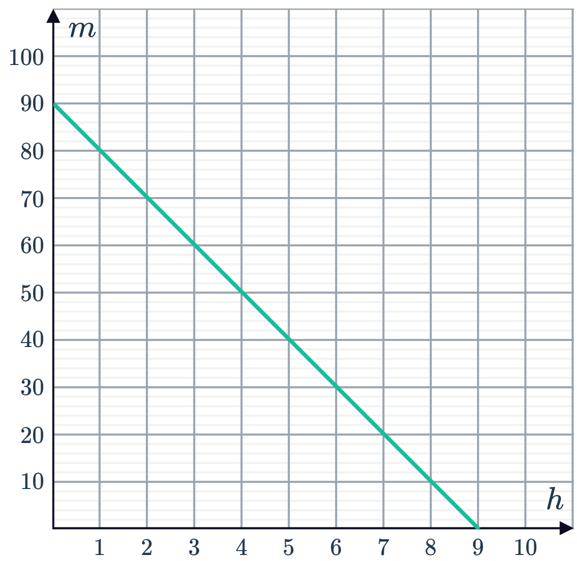


31 The number of hours spent watching TV each evening, h, is measured against the percentage results, m, achieved in the Economics exam.

a Find the y-intercept and interpret its meaning.

b Does the interpretation in the previous part make sense in this context?

c Would it be reasonable to use the line of best fit to predict average score of students who watch more than 10 hours of TV?



32 The average number of pages read to a ch[redacted]ach day and the child’s growing vocabulary are measured.

Pages read per day	25	27	29	3	13	31	18	29	29	5
Total vocabulary	402	440	467	76	220	487	295	457	460	106

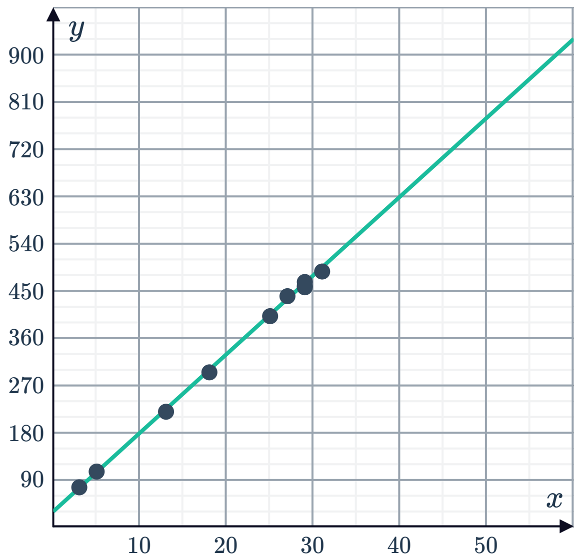
The data has been graphed along with a line of best fit.

a State the y-intercept.

b Interpret the meaning of the y-intercept.

c Does the interpretation in the previous part make sense in this context?

d Could we use the line of best fit to predict a child's vocabulary if they read more than 500 pages per day?



33 Concern over student use of the social media app SnappyChatty leads to a study of student scores in Mathematics versus minutes spent using the app.

Time (minutes)	292	153	354	253	11	42	195	7	162	254
Score	26	63	13	37	97	89	51	98	59	36

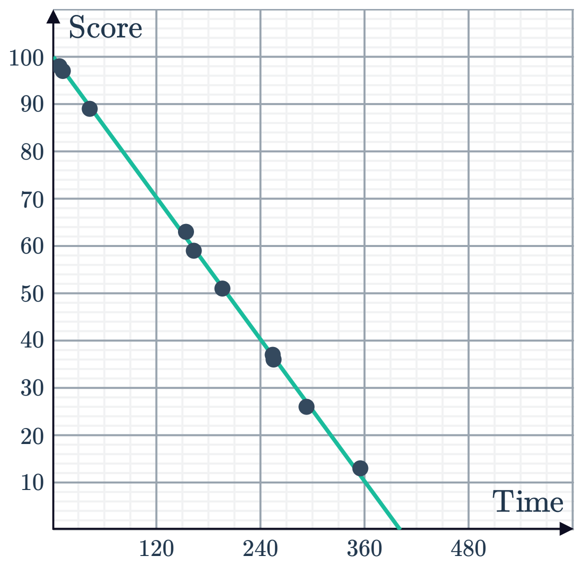
The data has been graphed along with a line of best fit.

a State the y-intercept, correct to the nearest ten.

b Interpret the meaning of the y-intercept.

c Does the interpretation in the previous part make sense in this context?

d Could we use the line of best fit to predict a student's score if they spent more than 400 minutes using SnappyChatty?



34 The table shows the price of various used [redacted]a Camrys.

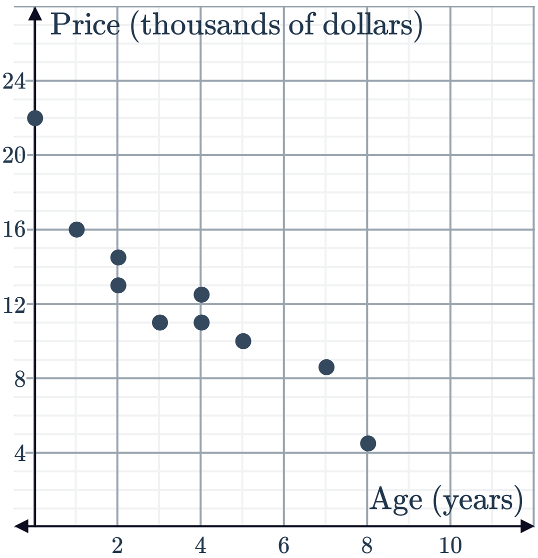
Age (years)	1	2	0	5	7	4	3	4	8	2
Price (thousands of dollars)	16	13	21.99	10	8.6	12.5	11	11	4.5	14.5

a Construct the line of best fit.

b Find the y-intercept of the line of best fit.

c What does the y-intercept represent?

d At what age would a Camry be worth $\$0$?



35 The following scattergraph shows the value of various 4 bedroom, 2 bathroom homes in a new suburb:

a State the y-intercept.

b Explain what the y-intercept represents.

c Does the interpretation in the previous part make sense in this context? Explain your answer.

d Use the graph to estimate the age of a house with an average value of $\$2\,000\,000$. Give your answer in years to one decimal place.

e Use the graph to estimate the average price of a house with age 3 years.